

2021

Q No 02:

Give short answers of the following:

(i) Write the Problems of Estimation?

In Statistical inference, Estimation involves making inferences about population parameters based on sample data. However, Estimation can encounter several problems or challenges.

1. Bias:

It occurs when the estimated value consistently deviates from the true population parameter. This can result from flaws in the sampling method or the estimation procedure.

2. Variance

Variance reflects the extent to which estimates vary across different samples. High variance indicates that the estimate is sensitive to the specific data used, which can lead to instability in inference and less confidence in the precision of the estimate.

3. Efficiency

Efficiency concerns the precision of an estimator. An efficient estimator achieves lower variance among unbiased estimators for a given sample size, indicating better precision in estimating the parameter.

4. Consistency

Consistency implies that as the sample size increase, the estimator converges to the true parameter value. Inconsistent Estimators fail to converge to the true parameter even with increasing sample size.

5. Robustness

Robustness refers to the ability of an estimator to perform well, even when underlying assumptions are violated or in the presence of outliers or influential data points.

6. Confidence Intervals:

Construction of intervals to quantify the uncertainty around the estimated parameter. Issues such as coverage probability and interval width are important considerations.

7. Assumptions:

Many estimation techniques rely on specific assumptions about the underlying pop distribution or relationship b/w variables. Violation of these assumptions can lead to biased or inefficient estimates.

According to these problems often require careful consideration of the estimation method, SS, robustness checks, and sensitivity analysis to assess the reliability and validity of the estimated parameters.

differentiate between:

- (a) Asymptotic unbiasedness and consistency
- (b) Efficiency and Mean square consistency

(a) Asymptotic unbiasedness and Consistency

Asymptotic unbiasedness and consistency are properties of estimation, while both properties deal with the behavior of estimators as the sample size increases. The key differences b/w asymptotic unbiasedness and consistency are as follows:

Asymptotic unbiasedness

1. definition:

It refers to the property that the expected value of the estimator equals to the true parameter value as the sample size approaches infinity.

2. focus:

It focuses on the expected value of the estimator.

Consistency

1. definition:

It refers to the property that the estimator converges in probability to the true parameter value as the sample size approaches infinity.

2. focus:

focuses on the behavior of the estimator itself, indicating its convergence to the true parameter value.

o. differentiate between:

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- (b) Efficiency and Mean square consistency

(a) Asymptotic unbiasedness and Consistency

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focuses on the behavior of the estimator itself, indicating its convergence to the true parameter value.

Asymptotic unbiasedness

3. Single vs Multiple Samples:

It guarantees unbiasedness in the long run over many samples.

4. Ease of interpretation

It's easier to interpret since it directly addresses the bias of the estimator.

5. Degree of convergence

It guarantees that the expected value of the estimator equals the true parameter value. but it does not necessarily imply that the estimator itself converges to the true value.

6. Robustness

Even in the presence of the certain robustness concerns such as outliers or model misspecifications, asymptotic unbiasedness holds.

Consistency:

3. Single vs Multiple Samples:

It guarantees the estimator accuracy for a single, sufficiently large sample.

4. Ease of interpretation:

It's sometimes less intuitive to interpret since it focuses on the convergence of the estimator.

5. Degree of convergence:

Consistency guarantees that the estimator converges in probability to the true parameter value, meaning that the estimator becomes arbitrarily close to the true value as the sample size increases.

6. Robustness

Consistency insures that the estimator converges in probability to the true parameter value as SS grows indefinitely, regardless of robustness concerns.

Efficiency and Meansquare Consistency:

Efficiency refers to how well an estimator utilizes the available information to estimate a parameter. It's about getting the most accurate estimate with the least amount of error.

An unbiased estimator is defined to be efficient estimator if the variance of its sampling distribution is smaller than that of the sampling distribution of any other unbiased estimator of the same parameter.

Let T_1 and T_2 be two unbiased estimators of the same parameter θ , Then T_1 is said to be more efficient estimator than T_2 , if:

$$V(T_1) < V(T_2)$$

Relative efficiency of T_1 compared with T_2 is:

$$R.E = \frac{Var(T_2)}{Var(T_1)} = 1 \quad \text{equally efficient}$$

$$R.E = \frac{Var(T_2)}{Var(T_1)} > 1 \quad T_1 \text{ is more Efficient}$$

$$R.E = \frac{Var(T_2)}{Var(T_1)} < 1 \quad T_2 \text{ is more efficient}$$

An unbiased estimator whose variance coincides with Cramer Rao lower Bound is called most efficient Estimator.

Mean Square Consistency:

Mean square consistency, is about how close an estimator gets to the true value of a parameter as the sample size increases to infinity, it's a property that ensures the estimator converges in probability to the true parameter value.

An estimator is said to be consistent in mean square of the population parameter θ if and only if,

$$E(\hat{\theta} - \theta)^2 \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$E(\hat{\theta} - \theta)^2$ is called the mean square Error (MSE) of the estimator ($\hat{\theta}$).

MSE can be defined as

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

$$= E(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2$$

$$= E(\hat{\theta} - E(\hat{\theta}))^2 + E(E(\hat{\theta}) - \theta)^2 + 2(E(\hat{\theta} - E(\hat{\theta})))(E(\hat{\theta}) - \theta)$$

$$\because E(\hat{\theta} - E(\hat{\theta})) = 0;$$

$$MSE(\hat{\theta}) = E(\hat{\theta} - E(\hat{\theta}))^2 + E(E(\hat{\theta}) - \theta)^2 + 0$$

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias})^2$$

For unbiased estimator,

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}).$$

$\hat{\theta}$ is said to be consistent in Mean square if

$$\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\lim_{n \rightarrow \infty} \text{bias} = 0 \quad \text{and} \quad \text{Var}(\hat{\theta}) \rightarrow 0$$

$E(\hat{\theta} - \theta)^2 \rightarrow 0$ is an estimator which is consistent in mean square.

for Biased estimators we compute MSE instead of variance.

Comparison of consistency and efficiency.

When consistency is measured in the form of the mean square consistency then it is approximately equal to the efficiency.

$$V(\hat{\theta}) \rightarrow 0 \text{ as } n \rightarrow \infty$$

(iii) How would you explain the Joint sufficiency?

Joint sufficiency in statistical inference, refers to a situation where a set of random variables contains all the information about the parameters of interest that is contained in any one of the variables individually.

In simpler terms, if you have a group of variables taken together, they provide all the relevant information about the parameters you are trying to estimate. This can simplify statistical analyses by reducing the number of variables needed to make accurate predictions or estimations.

iv. differentiate b/w location invariant and scale invariant estimators.

location invariant estimators also known as shift-invariant estimators, are statistical estimators whose values does not change, under translations or shifts of the underlying distribution. In other words if you add a constant value to all data points in a sample, the estimate remains the same.

Scale invariant estimators, on the other hand are statistical estimators whose values does not change under changes in scale or variance of the underlying distribution. If you multiply all data points in a sample by a constant value, the estimate remains the same.

For example the sample mean and median are location-invariant estimators because they do not change, if you add or subtract a constant from each data point. the sample median absolute deviation and coefficient of variation are scale-invariant estimators because they do not change, if you multiply each data point by a constant.

v). What do you mean by completeness.

Completeness refers to the property of a statistic or a set of statistics available in a sample about the parameters of interest.

formally let $T(X)$ be a statistic based on a random sample $X_1, X_2, \dots, X_n$ from a distribution with parameter θ . The statistic $T(X)$ is said to be complete if, for every function g , the condition.

$$E[g(T)] = 0 \quad ; \quad \text{for all } \theta \Rightarrow$$

$$g(T) = 0 \quad \text{almost surely holds,}$$

where E denotes the expected value.

In simpler terms, if the expected value of any function of the statistic $T(X)$ is zero for all possible values of the parameter θ , then that function must be zero almost surely. This means that $T(X)$ captures all information in the data about θ , making it a complete statistic.

Completeness is an important property in statistical theory because it ensures that there is no nontrivial unbiased estimator for certain functions of the parameter. In practical terms, completeness helps in finding the best estimators for parameters by ruling out certain types of estimators that cannot be improved upon.

Q NO 03:-

Let $Y_1 < Y_2 < Y_3$ be the order statistics of random sample of size 3 from uniform distribution over the interval $[0, \theta]$. Check unbiasedness of three statistics. How they can be made to be unbiased if some or all of these statistics are biased.

Solution:

→ The p.d.f of uniform distribution is,

$$f(y) = \frac{1}{b-a} \Rightarrow \frac{1}{\theta-0} = \frac{1}{\theta} \quad [0, \theta]$$

→ The distribution function will be;

$$F(y) = \int_a^y \frac{1}{\theta} dy \Rightarrow \frac{1}{\theta} \left| y \right|_0^y \Rightarrow \frac{1}{\theta} (y-0)$$

$$F(y) = \frac{y}{\theta}$$

→ The distribution of order statistic Y_r^n is

$$g(Y_r^n) = \frac{n!}{(r-1)!(n-r)!} [F'(Y_r^n)]^{r-1} [1-F(Y_r^n)]^{n-r} dF(Y_r^n)$$

→ Now distribution of Y_1 of random sample $n=3$, $r=1$

$$g(Y_1) = \frac{3!}{(1-1)!(3-1)!} (F'(Y_1))^{1-1} [1-F(Y_1)]^{3-1} f(Y_1)$$

$$= \frac{3 \cdot 2}{2} \left[(1) \left(1 - \frac{Y_1}{\theta} \right)^2 \right] \frac{1}{\theta}$$

$$g(Y_1) = \frac{3}{\theta} \left[1 - \frac{Y_1}{\theta} \right]^2$$

Now distribution of Y_2 of random sample $n=3, r=2$

$$g(Y_2) = \frac{3!}{(2-1)!(3-2)!} [F(Y_2)]^{2-1} [1-F(Y_2)]^{3-2} f(Y_2)$$

$$= \frac{3 \times 2}{(1)(1)} \left[\frac{Y_2}{\theta} \right] \left[1 - \frac{Y_2}{\theta} \right] \left(\frac{1}{\theta} \right)$$

$$g(Y_2) = \frac{6}{\theta} \left[\frac{Y_2}{\theta} \right] \left[1 - \frac{Y_2}{\theta} \right]$$

Now distribution of Y_3 of random sample $n=3, r=3$:

$$g(Y_3) = \frac{3!}{(3-1)!(3-3)!} [F(Y_3)]^{3-1} [1-F(Y_3)]^{3-3} f(Y_3)$$

$$= \frac{3 \cdot 2}{2(1)} \left[\frac{Y_3}{\theta} \right]^2 (1) \left(\frac{1}{\theta} \right)$$

$$g(Y_3) = \frac{3}{\theta} \left(\frac{Y_3}{\theta} \right)^2$$

To check unbiasedness:

$$E(Y_1) = \int_0^{\theta} Y_1 g(Y_1) dY_1 \Rightarrow \frac{3}{\theta} \int_0^{\theta} Y_1 \left(1 - \frac{Y_1}{\theta} \right)^2 dY_1$$

$$= \frac{3}{\theta} \int_0^{\theta} Y_1 \left(1 - \frac{Y_1}{\theta} \right)^2 dY_1$$

let

$$Z = 1 - \frac{Y_1}{\theta} \Rightarrow \frac{Y_1}{\theta} = 1 - Z \Rightarrow Y_1 = \theta - \theta Z$$

$$Y_1 = \theta(1 - Z)$$

$$dY_1 = -\theta dZ$$

$$\text{when } Y_1 = \theta; \text{ then } Z = 1 - \frac{\theta}{\theta} \Rightarrow 1 - 1 = 0$$

$$Y_1 = 0 \text{ then } Z = 1 - \frac{0}{\theta} \Rightarrow 1 - 0 = 1$$

$$E(Y) = \frac{3}{\theta} \int_0^{\theta} y_1 \left(1 - \frac{y_1}{\theta}\right)^2 dy_1$$

$$E(Y_1) = \frac{3}{\theta} \int_0^{\theta} \theta(1-z) (z)^2 \theta dz$$

$$= \frac{3\theta^2}{\theta} \int_0^1 (1-z) z^2 dz$$

$$= \frac{3\theta^2}{\theta} \int_0^1 (z^2 - z^3) dz$$

$$= 3\theta \left[\frac{z^3}{3} - \frac{z^4}{4} \right]$$

$$= 3\theta \left[\frac{(1)^3 - (0)^3}{3} - \frac{(1)^4 - (0)^4}{4} \right]$$

$$= 3\theta \left(\frac{1}{3} - \frac{1}{4} \right) \Rightarrow 3\theta \left(\frac{4-3}{12} \right)$$

$$E(Y_1) = \frac{\theta}{4}$$

$\Rightarrow E(Y_1) \neq \theta$ hence Y_1 is not an unbiased estimator of θ .

$$E(Y_2) = \int_0^{\theta} y_2 g(y_2) dy_2$$

$$= \frac{6}{\theta} \int_0^{\theta} y_2 \left(\frac{y_2}{\theta} \right) \left(1 - \frac{y_2}{\theta} \right) dy_2$$

let;

$$z = 1 - \frac{y_2}{\theta} ; \quad \frac{y_2}{\theta} = 1 - z$$

$$y_2 = \theta - \theta z \Rightarrow \theta(1-z)$$

$$dy_2 = -\theta dz$$

$$17 \quad y_2 = 0 \quad \text{then} \quad z = 1 - \frac{0}{\theta} \Rightarrow 1.$$

$$17 \quad y_2 = \theta \quad \text{then} \quad z = 1 - \frac{\theta}{\theta} \Rightarrow 1 - 1 = 0$$

$$E(\hat{y}_2) = \frac{6}{\theta} \int_0^{\theta} y_2 \left(\frac{y_2}{\theta} \right) \left(1 - \frac{y_2}{\theta} \right) dy_2$$

$$= \frac{6}{\theta} \int_0^{\theta} \theta(1-z)(1-z)(z-\theta dz)$$

$$= \frac{6\theta z^2}{\theta} \int_0^1 (1-z)^2 z dz$$

$$= 6\theta \int_0^1 (z^2 + 1 - 2z) z dz$$

$$= 6\theta \int_0^1 (z^3 + z - 2z^2) dz$$

$$= 6\theta \left[\frac{z^4}{4} + \frac{z^2}{2} - 2 \frac{z^3}{3} \right]$$

$$= 6\theta \left[\frac{(1)^4 - (0)^4}{4} + \frac{(1)^2 - (0)^2}{2} - 2 \frac{(1^3 - 0^3)}{3} \right]$$

$$= 6\theta \left[\frac{1}{4} + \frac{1}{2} - \frac{2}{3} \right]$$

$$= \frac{6\theta}{12}$$

$$E(\hat{y}_2) = \theta/2$$

$\Rightarrow E(\hat{y}_2) \neq \theta$ hence $\hat{y}_2$ is not an unbiased estimator of θ .

$$E(Y_3) = \int_0^{\theta} Y_3 g(Y_3) dY_3$$

$$= \frac{3}{\theta} \int_0^{\theta} Y_3 \left(\frac{Y_3}{\theta}\right)^2 dY_3$$

$$\frac{Y_3}{\theta} = Z ; \quad Y_3 = \theta Z.$$

$$dY_3 = \theta dZ.$$

Limits:

$$\text{If } Y_3 = \theta \text{ then } \left(Z = \frac{\theta}{\theta} = 1\right) \Rightarrow Z = 1.$$

$$\text{If } Y_3 = 0 \text{ then } \left(Z = \frac{0}{\theta} = 0\right) \Rightarrow Z = 0.$$

$$E(Y_3) = \frac{3}{\theta} \int_0^1 (\theta Z) Z^2 (\theta dZ)$$

$$= \frac{3\theta^2}{\theta} \int_0^1 Z^3 dZ.$$

$$= 3\theta \left[\frac{Z^4}{4} \right]_0^1$$

$$= 3\theta \left[\frac{1^4 - 0^4}{4} \right]$$

$$E(Y_3) = \frac{3}{4} \theta.$$

$\Rightarrow E(Y_3) \neq \theta$ hence Y_3 is also not an unbiased estimator of θ .

How they can become unbiased:

$$4E(Y_1) = 4\left(\frac{\theta}{4}\right) = \theta.$$

$$2E(Y_2) = 2\left(\frac{\theta}{2}\right) = \theta$$

$$\frac{4}{3}E(Y_3) = \frac{4}{3}\left(\frac{3}{4}\theta\right) = \theta.$$

Q No 04:-

state and prove uniqueness
Theorem for efficiency:

Solution

A minimum variance unbiased estimator is unique in the sense that T_1 and T_2 are MVU estimators for $T(\theta)$ then, $T_1 = T_2$ almost surely.

$\Rightarrow$ As T_1 and T_2 both are unbiased estimators of $T(\theta)$, so;

$$E(T_1) = E(T_2) = T(\theta)$$

$\Rightarrow$ also T_1 and T_2 are minimum variance estimators

So; $V(T_1) = V(T_2) = V$ All θ related to number line
 $V \in \theta \in \theta$

$\Rightarrow$ Now let's consider another unbiased estimator of $T(\theta)$.

$$T = \frac{1}{2} (T_1 + T_2)$$

this estimator T is unbiased:

Mean $E(T) = \frac{1}{2} (E(T_1) + E(T_2))$

$$= \frac{1}{2} (T(\theta) + T(\theta))$$

$$= T(\theta)$$

$$E(T) = T(\theta) \quad \text{proved:}$$

Variances of T :

$$V(T) = \text{var} \left(\frac{T_1 + T_2}{2} \right)$$

$$= \frac{1}{4} \left(\text{var}(T_1) + \text{var}(T_2) + 2\text{cov}(T_1, T_2) \right)$$

$$V(T) = \frac{1}{4} \left(2(\text{var} T_1) + 2\text{cov}(T_1, T_2) \right) \quad \therefore \text{var}(T_1) = \text{var}(T_2)$$

Now we find the value of $\text{cov}(T_1, T_2)$.

⇒ The correlation coefficient b/w T_1 & T_2 is:

$$\rho = \frac{\text{cov}(T_1, T_2)}{\sqrt{\text{var}(T_1) \text{var}(T_2)}}$$

$$= \frac{\text{cov}(T_1, T_2)}{\sqrt{(\text{var}(T_1))^2}}$$

$$\because \text{var}(T_1) = \text{var}(T_2)$$

$$\rho = \frac{\text{cov}(T_1, T_2)}{\text{var}(T_1)}$$

$$\text{cov}(T_1, T_2) = \rho (\text{var}(T_1))$$

Now:

$$\text{var}(T) = \frac{1}{4} [2\text{var}(T) + 2\text{cov}(T_1, T_2)]$$

$$= \frac{1}{4} [2\text{var}(T_1) + 2\rho(\text{var}(T_1))]$$

$$= \frac{1}{4} \cdot 2\text{var}(T_1) [1 + \rho]$$

$$\text{var}(T) = \frac{1}{2} (\text{var}(T_1)) (1 + \rho)$$

PUACP

Since T_1 and T_2 are Minimum var unbiased Estimator:

$$V(T) \geq V(T_1)$$

$$\because V(T_1) = V(T_2)$$

$$\frac{1}{2} (\text{var}(T_1) (1 + \rho)) \geq \text{var}(T_1)$$

$$\text{var}(T_1) (1 + \rho) \geq 2 \text{var}(T_1)$$

$$(1 + \rho) \geq 2$$

$$\rho \geq 2 - 1$$

$$\rho \geq 1$$

But as we know that $|P| \leq 1$, or the minimum value of ρ is 1.

Hence we consider $\rho = 1$; Then T_1 and T_2 must have a linear relationship of the form,

$$T_1 = \alpha + \beta T_2 \quad \text{--- (1)}$$

Taking expectation:

$$E(T_1) = \alpha + \beta E(T_2)$$

$$T(0) = \alpha + \beta T(0) \quad \text{--- (2)}$$

Applying variance on eq (1),

$$\text{var}(T_1) = \text{var}(\alpha + \beta T_2)$$

$$\text{var}(T_1) = 0 + \beta^2 \text{var}(T_2)$$

$$\text{var}(T_1) = \beta^2 \text{var}(T_2) \quad ; \quad \text{var}(T_1) = \text{var}(T_2)$$

$$\beta^2 = 1$$

$$\beta = \pm 1$$

here $\beta = 1$ because;

$$\text{corr}(T_1, T_2) = 1$$

$\Rightarrow$ Let's put $\beta = 1$ in eq (2)

$$T(0) = \alpha + \beta(T(0))$$

$$T(0) = \alpha + (1)T(0)$$

$$T(0) = \alpha + T(0) \quad \therefore \alpha = 0$$

$$T(0) - T(0) = \alpha$$

$$\alpha = 0$$

$\Rightarrow$ Now put $\alpha = 0$ and $\beta = 1$ in (1).

$$T_1 = 0 + (1)T_2$$

$$\boxed{T_1 = T_2} \quad \text{Hence proved}$$

Q No 05:

Let $t = x_1 + x_2 + x_3$; $k = x_1 + 2x_2 + x_3$
be two statistics where x_i follow Bernoulli distribution, Check whether or not the 't' and 'k' are sufficient by using conditional probability approach.

Sol

⇒ we have given that;

$$t = x_1 + x_2 + x_3 ; \quad k = x_1 + 2x_2 + x_3$$

⇒ The p.d.f of Bernoulli dist is given by;

$$f(x_i) = \theta^{x_i} (1-\theta)^{1-x_i} \quad \because x_i = 0, 1$$

Sample space $(2)^3 = 8$

	x_1	x_2	x_3	t	k	probability
1	0	0	0	0	0	$\theta^{0+0+0} (1-\theta)^{3-(0+0+0)} = (1-\theta)^3$
2	0	0	1	1	1	$\theta^{0+0+1} (1-\theta)^{3-(0+0+1)} = \theta (1-\theta)^2$
3	0	1	0	1	2	$\theta^{0+1+0} (1-\theta)^{3-(0+1+0)} = \theta (1-\theta)^2$
4	0	1	1	2	3	$\theta^{0+1+1} (1-\theta)^{3-(0+1+1)} = \theta^2 (1-\theta)$
5	1	0	0	1	1	$\theta^{1+0+0} (1-\theta)^{3-(1+0+0)} = \theta (1-\theta)^2$
6	1	0	1	2	2	$\theta^{1+0+1} (1-\theta)^{3-(1+0+1)} = \theta^2 (1-\theta)$
7	1	1	0	2	3	$\theta^{1+1+0} (1-\theta)^{3-(1+1+0)} = \theta^2 (1-\theta)$
8	1	1	1	3	4	$\theta^{1+1+1} (1-\theta)^{3-(1+1+1)} = \theta^3 (1)$

$$L(\theta; x) = \prod_{i=1}^3 f(x_i; \theta)$$

$$= \prod_{i=1}^3 \theta^{x_i} (1-\theta)^{1-x_i}$$

$$= \theta^{x_1 + x_2 + x_3} (1-\theta)^{1-x_1} \cdot \theta^{x_2} (1-\theta)^{1-x_2} \cdot \theta^{x_3} (1-\theta)^{1-x_3}$$

$$= \theta^{x_1 + x_2 + x_3} (1-\theta)^{1-x_1 + 1-x_2 + 1-x_3} \Rightarrow \theta^{x_1 + x_2 + x_3} (1-\theta)^{3-(x_1 + x_2 + x_3)}$$

$$L(\theta, x) = \theta^{x_1+x_2+x_3} (1-\theta)^{3-(x_1+x_2+x_3)}$$

Now we compute conditional probabilities for each T and k ,

for $t=0$

$$P(x_1=0, x_2=0, x_3=0 | t=0) = \frac{P(x_1=0, x_2=0, x_3=0)}{P(T=0) \Rightarrow P(x_1=0, x_2=0, x_3=0)}$$

↓
∴ (The points where t is = 0)

$$= \frac{(1-\theta)^3}{(1-\theta)^3}$$

$$P(0,0,0 | t=0) = 1 \quad \text{independent of } \theta \quad \text{sufficient } t=0:$$

For $k=0$;

$$P(x_1=0, x_2=0, x_3=0 | k=0) = \frac{P(x_1=0, x_2=0, x_3=0)}{P(k=0) \Rightarrow P(x_1=0, x_2=0, x_3=0)}$$

$$= \frac{(1-\theta)^3}{(1-\theta)^3}$$

$$P(0,0,0 | k=0) = 1$$

independent of θ so sufficient $\Rightarrow k=0$

For $t=1$ possible outcomes are $(0,0,1), (0,1,0), (1,0,0)$.

we can have;

$$P(x_1=0, x_2=0, x_3=1) = P(x_1=0, x_2=0, x_3=1) = P(x_1=1, x_2=0, x_3=0)$$

$$\theta(1-\theta)^2 = \theta(1-\theta)^2 = \theta(1-\theta)^2$$

$$P(t=1) \Rightarrow \theta(1-\theta)^2 + \theta(1-\theta)^2 + \theta(1-\theta)^2$$

$$P(t=1) = 3\theta(1-\theta)^2$$

$$P(1,0,0|t=1) \Rightarrow P(0,1,0|t=1) \Rightarrow P(0,0,1|t=1) \\ = \frac{\theta(1-\theta)^2}{3\theta(1-\theta)^2}$$

$$P(1,0,0|t=1) = P(0,1,0|t=1) = P(0,0,1|t=1) = \frac{1}{3}$$

sufficient $\Rightarrow t=1$;

for $k=1$, possibilities are $(0,0,1)$ & $(1,0,0)$.

$$P(X_1=0, X_2=0, X_3=1) = P(X_1=1, X_2=0, X_3=0)$$

$$\theta(1-\theta)^2 = \theta(1-\theta)^2$$

$$P(k=1) = \theta(1-\theta)^2 + \theta(1-\theta)^2$$

$$P(k=1) = 2\theta(1-\theta)^2$$

$$P(0,0,1|k=1) = P(1,0,0|k=1) = \frac{\theta(1-\theta)^2}{2\theta(1-\theta)^2}$$

$$P(0,0,1|k=1) = P(1,0,0|k=1) = \frac{1}{2}$$

sufficient $\Rightarrow k=1$ as;

both t and k at '1' are independent of θ ,

for $t=2$: the possibilities are $(0,1,1)$ $(1,0,1)$ $(1,1,0)$

$$P(X_1=0, X_2=1, X_3=1) = P(X_1=1, X_2=0, X_3=1) = P(X_1=1, X_2=1, X_3=0)$$

$$\theta^2(1-\theta) = \theta^2(1-\theta) = \theta^2(1-\theta)$$

$$P(t=2) = \theta^2(1-\theta) + \theta^2(1-\theta) + \theta^2(1-\theta)$$

$$P(t=2) = 3\theta^2(1-\theta)$$

now,

$$P(0,1,1|t=2) = P(1,0,1|t=2) = P(1,1,0|t=2) = \frac{\theta^2(1-\theta)}{3\theta^2(1-\theta)} \\ = \frac{1}{3}$$

sufficient $t=2$

for $k=2$: the possibilities are $(0,1,0)$ $(1,0,1)$
 $P(X_1=0, X_2=1, X_3=0) = P(X_1=1, X_2=0, X_3=1) =$

$$0(1-0)^2 \neq 0^2(1-0)$$

$$P(k=2) = 0(1-0)^2 + 0^2(1-0) \Rightarrow$$

$$P(0,1,0|k=2) \neq P(1,0,1|k=2) = \text{unequal that's why}$$

for $k=2$ is not sufficient due to H_0 's conditional distribution is not independent of θ .

For $t=3$: $(1,1,1)$.

$$P(X_1=1, X_2=1, X_3=1) = \theta^3.$$

$$P(T=3) = P(X_1=1, X_2=1, X_3=1) = \theta^3.$$

$$P(1,1,1|t=3) = \frac{\theta^3}{\theta^3} = 1.$$

Independent of θ , $t=3$ is sufficient

For $k=3$: $(0,1,1)$ $(1,1,1)$

$$P(X_1=0, X_2=1, X_3=1) = P(X_1=1, X_2=1, X_3=1)$$

$$0^2(1-0) = 0^2(1-0)$$

$$P(k=3) = 0^2(1-0) + 0^2(1-0)$$

$$= 2 \cdot 0^2(1-0)$$

$$P(0,1,1|k=3) = P(1,1,1|k=3) = \frac{0^2(1-0)}{2 \cdot 0^2(1-0)} = \frac{1}{2}$$

Independent of θ hence $k=3$ is sufficient.

for $k=4, [(1, 1, 1)]$.

$$P(X_1=0, X_2=1, X_3=1) = 0^3.$$

$$P(k=4) = 0^3$$

$$P(1, 1, 1 | k=4) = \frac{0^3}{0^3} = 1.$$

independent of θ hence $k=4$ is also
sufficient estimator.

PUACP